

1. a.  $\Pi_{ID, name} (\sigma_{dept-name = 42})$

employee : pid  
works : pid  
company : cid

a.  $\Pi_{pid, person\ name} (\sigma_{company\ name \neq "Big\ Bank"} (works))$

b.  $A \leftarrow works$

$\Pi_{works.pid, works.person\ name} (A \bowtie_{A.salary = works.salary} works)$

3. a.  $\Pi_{loan-number} (\sigma_{amount > 10000} (loan))$

$\Pi_{ID} 3. a. \Pi_{loan-number} (\sigma_{amount > 10000} (loan))$

b.  $\Pi_{ID} (\sigma_{account-number > 6000} (depositor))$

$\Pi_{depositor.ID} (\sigma_{account \bowtie_{account.account-number = depositor.account-number} account, balance > 6000})$

c.  $\Pi_{depositor.ID} (\sigma_{account.balance > 6000 \wedge account.branch-name = "Uptown"} (account \bowtie_{account.account-number = depositor.account-number} depositor))$

4. 数据库中空值主要用来表示值不存在和值未知

d. a.  $\pi_{ID, name} (\sigma_{dept-name = \text{物理}} (\text{instructor}))$

b.  $\pi_{instructor.ID, instructor.name} (\sigma_{(department \bowtie instructor) \wedge \text{building} = \text{"Watson"} \wedge department.dept-name = instructor.dept-name})$

c.

$A \leftarrow \text{student} \bowtie \text{takes}$   
 $\text{student.ID} = \text{takes.ID}$   
 $\text{course} \bowtie A$   
 $\text{course-id} = A.\text{course-id}$

c.

$A \leftarrow \text{student} \bowtie \text{takes}$   
 $\text{student.ID} = \text{takes.ID}$

$B \leftarrow \text{course} \bowtie A$   
 $\text{course.course-id} = A.\text{course-id}$

$\pi_{B.name, B.student.ID} (\sigma_{\text{title} = \text{"Comp.Sci."} (B)})$

d.  $A \leftarrow \text{student} \bowtie \text{takes}$   
 $\text{student.ID} = \text{takes.ID}$

$\pi_{A.student.ID, name} (\sigma_{\text{year} = 2018 (A)})$

e. 记 d 结果为 D

$\pi_{ID, name} (\text{student}) - D$